Chapter 6: Expectations

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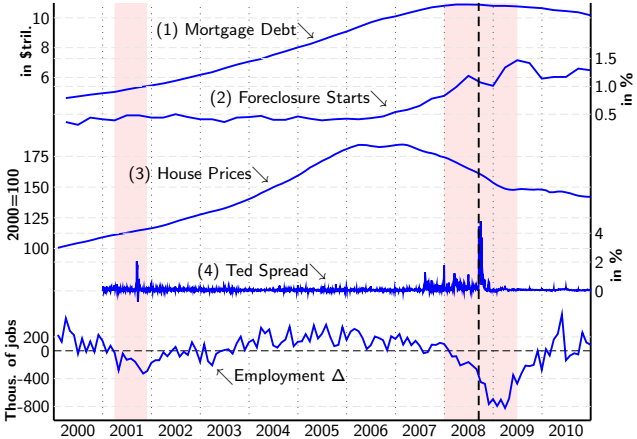
Course Overview



- (1) Debt Growth
- (2) Default
- (3) House Prices
- (4) Financial Crisis
- (5) Institutions
- (6) Expectations

Harrison and Kreps (1978)

- Consider an asset x which pays off x in state s and zero otherwise.
- General proposition:
  - All else equal, the more likely I believe state s to occur, the more I will pay for asset x.
- From asset prices we can back out probabilities.
- Harrison and Kreps (1978) show an example of the opposite

- 2 investor classes $a = 1, 2$
  - continuum of agents of each type
  - linear utility
  - infinite wealth
  - discount rate $\gamma = 0.75$
  - no short sales
- Discrete time $t = 0, ..., \infty$
- One asset
  - Two states in each period
  - Pays 0 in state 0 and 1 in state 1
- Markov chain:

$$\begin{array}{l} d(0) = 0 \\ d(1) = 1 \end{array} \qquad Q^a \begin{bmatrix} q^a(0,0) & q^a(0,1) \\ q^a(1,0) & q^a(1,1) \end{bmatrix}$$

Harrison and Kreps (1978) II

- Suppose asset will pay off 1 with probability q and 0 with probability 0 each period until it pays and then never again

$$\begin{aligned} V(s) &= \textstyle\sum_{t=s+1}^{\infty}(\gamma q)^{t-s}) \\ &= \textstyle\sum_{t=s}^{\infty}(\gamma q)^{t-s} - 1 \\ &= (1-\gamma q)^{-1} - 1 \end{aligned}$$

- For Markov chain

$$V\begin{pmatrix} 0 \\ 1 \end{pmatrix} = (I - \gamma Q^a)^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- agents 1 and 2

$$Q^1 = \begin{bmatrix} 1/2 & 1/2 \\ 2/3 & 1/3 \end{bmatrix} \quad Q^2 = \begin{bmatrix} 2/3 & 1/3 \\ 1/4 & 3/4 \end{bmatrix}$$

- Valuations:

$$v^1(0) = \tfrac{4}{3} = 1.33 \quad v^1(1) = \tfrac{11}{9} = 1.22$$
$$v^2(0) = \tfrac{16}{11} = 1.45 \quad v^2(1) = \tfrac{21}{11} = 1.91$$

1. Agent 2 always values the security more
   - Good state is very "persistent"
   - No short sales $\Rightarrow$ always holds
2. Agent 1 believes bad state is less persistent

Harrison and Kreps (1978) III

- Is $p(0) = v^2(0) = 1.45$ and $p(1) = v^2(1) = 1.91$ an equilibrium?
- No!
- Suppose Class 1 investors use the following strategy

$$\begin{cases} (1) & \text{Buy asset in state 0} \\ (2) & \text{Sell asset in state 1 for 1.91} \end{cases}$$

- Payoff

$$\begin{aligned} &= [0.75 \cdot (1/2) + (0.75 \cdot 1/2)^2 + ...](1 + 1.91) \\ &= [(1 - 0.75 \cdot (1/2))^{-1} - 1] \cdot 2.91 \\ &= 1.75 > 1.45 = v^2(0) \end{aligned}$$

- Is $p(0) = 1.75$ and $p(1) = 1.91$ an equilibrium?
- No!
- Suppose Class 2 investors use the following strategy

$$\begin{cases} (1) & \text{Buy asset in state 1} \\ (2) & \text{Sell asset in state 0 for 1.75} \end{cases}$$

- Payoff

$$\begin{aligned} &= [0.75 \cdot (\tfrac{3}{4} \cdot 1 + \tfrac{1}{4} \cdot 1.5) + +[0.75^2 \tfrac{3}{4} \cdot (\tfrac{3}{4} \cdot \\ &= 2.03 > 1.91 = v^2(1) \end{aligned}$$

- Can go on. Harrison and Kreps (1978) show equilibirium is $p(0) = 1.85$ and $p(1) = 2.08$

Harrison and Kreps (1978) IV

1. Type 1 are "optimists" in state 0 – bad state is nt persistent. Type 2 are optimists in state 0 – good state is persistent
2. Short sales are key == pessimists cannot pull price down.
3. Same argument (in continuous time) in Scheinkman and Xiong (2003)
4. Suppose there were naiive agents trying to infer probabilities from prices.
5. There is a mistake. $p * (1) = 2.08$ not 2.04
6. Market timing matters
7. EMH depends on everyone having same beliefs:

As we understand things, a subjectivist statement of [the EMH] implicitly requires the supposition that rational economic agents having the same information will arrive at the same subjective probability assessments. The [EMH] then says that although investors do not originally have the same information, prices reflect this information in such a way as to induce actions identical to those they would undertake if they have complete information. Thus the investors are effectively homogenous in an "efficient market."

Shleifer and Vishny (1997)

- Efficient markets hypothesis says:
  - Asset Prices should be unpredictable
  - You can't beat the market.
- Bubbles confound this:
  - For a while, people predictably beat the market.
  - But the inevitable fall is predictable also.
- Why doesn't arbitrage work?
- "Limits of Arbitrage." Shleifer and Vishny (1997).

Short-selling

- Stock trading for $100 a share
- You know it will trade for $150 next week.
- Arbitrage opportunity!
  - Borrow $100, buy the stock and sell it next week for $150.
- Suppose you know it will trade $50.
- Arbitrage opportunity
  - Need to short.
- Securities lending
  - Borrow the stock today
  - Promise to return the *stock* tomorrow.
- To make money:
  - Today:
    - Borrow *the stock*
    - Sell the stock for $100
    - Deposit the $100 in a bank account
  - Tomorrow:
    - Buy the stock for $50
    - Return *the stock* to the owner
- Make a profit of $50

$t = 1$	$t = 2$	$t = 3$

$P = 100$

Share position	$-\$100$
Bond position	$+\$100$
Net Position	$+\$0$

$P = 75$

Share position	$-\$75$
Bond position	$+\$105$
Net Position	$+\$30$

$P = 50$

Share position	$-\$50$
Bond position	$+\$110.25$
Net Position	$+\$60.25$

$P = 126$

Share position	$-\$126$
Bond position	$+\$105$
Net Position	$-\$21-\$21$

$P = 50$

Share position	$-\$50$
Bond position	$+\$110.25$
Net Position	$+\$60.25$

- Stock trading for \$100 today...
- but only \$50 at $t = 3$
- Arbitrage Opportunity!
- Short sell!

- Suppose that at $t = 2$
- Two possibilities
- Stock goes down to \$75
- Stock goes up to \$126
- Still \$50 at $t = 3$.

- Still an arbitrage opportunity
- But along one path
- You will be down \$21.
- Is that a problem?

Margin Requirements

- Securities lenders are nervous about this situation.
- If they wanted their loan back now, you would have to buy the stock for $126 but you only have $105.
- To prevent this, they insist on a "margin."
  - 20% margin requirement says
  - That you must put down 20% of the value of the stock you are shorting in case.
- Let's see how this would work.

$t = 1$	$t = 2$	$t = 3$

$P = 100$

Share position	$-\$100$
Bond position	$+\$120$
Net Position	$+\$20$

$P = 75$

Share position	$-\$75$
Bond position	$+\$126$
Net Position	$+\$51$

$P = 50$

Share position	$-\$50$
Bond position	$+\$132.3$
Net Position	$+\$82.3$

$P = 126$

Share position	$-\$126$
Bond position	$+\$126$
Net Position	$+\$0$

$P = 50$

Share position	$+\$0$
Bond position	$+\$0$
Net Position	$+\$0$

- Lender requires that you buy $120 of bonds
- You must now invest $20
- But along good path you still make money

- On the bad path
- Net position of zero..
- Why is that bad? No margin!
- No more short selling.

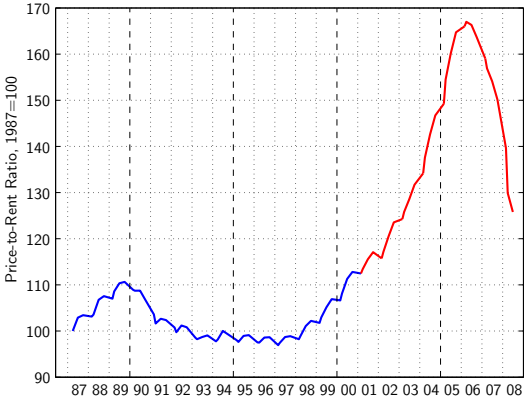
- Formerly a riskless bet is now risky.
- You invest $20
  - You can make $82.3 or lose it all.
- No arbitrage opportunity.

Example: Housing

- John Maynard Keynes said that:

"Market can stay irrational longer than you can stay solvent."

- In 2000s, Price-to-rent ratio went up and up
- And then it went up more!
- People who bet against housing in 2004 lost everything!



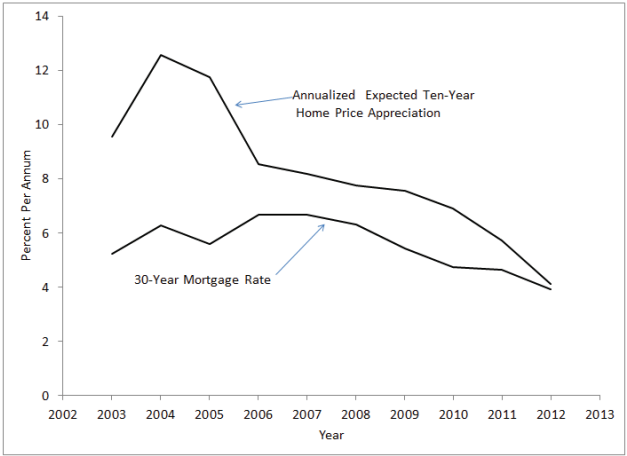
Lo and McKinlay (1990)

- Mean reversion in stocks in daily data
  - Stocks that do better than average today
  - Do worse than average tomorrow
- Can beat market by a tiny bit by creating a portfolio every day
  - Long all stocks that did worse than average
  - Short all stocks that did better than average

Year	Market Capitalization Deciles										All
	Smallest	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Largest	
	Average Daily Returns										
1995	3.57%	2.75%	1.94%	1.62%	1.07%	0.61%	0.21%	-0.01%	-0.02%	0.04%	1.38%
1996	3.58%	2.47%	1.82%	1.34%	0.84%	0.52%	0.19%	-0.11%	-0.04%	0.02%	1.17%
1997	2.83%	1.94%	1.34%	1.02%	0.62%	0.28%	0.04%	-0.12%	0.06%	0.14%	0.88%
1998	2.38%	1.45%	1.11%	0.62%	0.29%	0.03%	-0.04%	-0.12%	0.03%	0.10%	0.57%
1999	2.56%	1.41%	0.82%	0.38%	-0.01%	-0.11%	-0.21%	-0.35%	-0.01%	0.06%	0.44%
2000	2.58%	1.59%	0.92%	0.14%	0.03%	-0.02%	-0.14%	0.16%	0.00%	0.03%	0.44%
2001	2.15%	1.25%	0.57%	0.24%	-0.01%	0.06%	0.13%	-0.10%	-0.11%	-0.11%	0.31%
2002	1.67%	0.85%	0.53%	0.29%	0.28%	0.26%	0.28%	0.20%	0.11%	0.09%	0.45%
2003	1.00%	0.26%	-0.07%	0.04%	0.11%	0.20%	0.18%	0.15%	0.04%	0.05%	0.21%
2004	1.17%	0.48%	0.31%	0.38%	0.25%	0.29%	0.22%	0.15%	0.05%	-0.01%	0.37%
2005	1.05%	0.39%	0.13%	0.11%	0.09%	0.11%	0.05%	0.08%	0.01%	0.02%	0.26%
2006	0.86%	0.26%	0.11%	0.06%	0.05%	-0.02%	-0.02%	0.05%	0.06%	0.00%	0.15%
2007	0.57%	0.09%	0.08%	0.18%	0.16%	-0.08%	0.04%	-0.04%	0.00%	-0.04%	0.13%

Fact 13: Borrowers were optimistic about house prices

- "What have they been thinking? Home buyer behavior in hot and cold markets"
- *Brookings Papers on Economic Activity*, Fall 2012



NY Fed Survey of Consumer Expectations

- Online survey with roughly 1,200 respondents ("household heads"), staying in sample for up to 12 months (rotating panel).
  1. Monthly module: inflation expectations, labor market expectations, etc.
  2. Quarterly (special) modules: in this case on housing and mortgages
- Housing module fielded in Feb 2015; 87% of invited households participated

- Geographically representative of US
- Along many dimensions, average characteristics close to US average (e.g. age, income).
- 74.3% homeowners (vs. 63.7% homeownership rate according to Census)
- More educated: 55% have at least a BA, vs. 33% in overall population
  - Likely due to differential internet access / computer literacy
- Trim expectations / perceptions at 2% throughout

Survey Design

3 stages of main survey:

1. **Baseline Stage.** Respondents asked about:
   - Price of typical home in their zip code
   - Perceived change over *past year* and *past 5 years*
   - Expected change over *next year* and *next 5 years*
   - Hypothetical one-year investment scenario:
     allocate $1000 between risk-free savings account (2% interest) and housing market fund with return = local HPA.
2. **Treatment Stage** ($\sim$ 15 min later). Respondents randomized:
   - **T1:** Informed about change in local (mostly zip code) Zillow Home Value Index over 2014.
   - **T5:** Informed about change in Zillow Home Value Index over 2010-2014.
   - **Control:** No Zillow info

Survey Design (cont'd)

3. **Final Stage** (right after treatment). Respondents asked about:
   - Expected change in zip code HP over *next year* and *next 5 years*
   - One-year investment scenario:
     allocate $1000 between risk-free savings account (2% interest) and housing market fund with return = zip code HPA.
     Now <u>incentivized</u>: 2 randomly chosen respondents get paid out.

Follow-up survey: two months later (April 2015), respondents that are still in survey panel asked again about expected change in zip code HP over next year and next 5 years.

Descriptive Statistics – Baseline Stage

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	Full Sample	Control	T1	T5	p-value
Perceptions					
Past 1 year HP change	3.81	3.76	3.42	4.27	0.120
	(5.43)	(5.19)	(5.40)	(5.66)	
Past 5 year HP change	1.53	1.60	1.43	1.57	0.756
(annualized)	(3.04)	(2.85)	(3.10)	(3.18)	
Perception Gaps (= Zillow HPA − perceived HPA)					
1 year Perception Gap	1.38	1.42	1.43	1.28	0.954
	(3.83)	(3.72)	(3.79)	(3.96)	
5 year Perception Gap	-0.51	-0.80	-0.36	-0.39	0.286
(annualized)	(4.07)	(3.93)	(4.09)	(4.19)	
Expectations					
1 year E(HPA)	3.51	3.39	3.41	3.73	0.427
	(3.83)	(3.72)	(3.79)	(3.96)	
2-5 year E(HPA)	1.70	1.51	1.88	1.72	0.022

Empirical Momentum/Mean Reversion Patterns in HPA

Regressions using CoreLogic HPI 1976-2015:

	Survey Sample ($N = 753$)			National Sample ($N = 7133$)		
	Estimates	Percent Positive[a]	Percent Negative[a]	Estimates	Percent Positive[a]	Percent Negative[a]
1yr Δ on lagged 1yr Δ	0.53 (0.14)	91.2%	0.0%	0.53 (0.14)	89.7%	0.0%
2-5yr Δ on lagged 1yr Δ	0.03 (0.12)	8.1%	1.7%	0.03 (0.12)	8.8%	1.2%
1yr Δ on lagged 5yr Δ	0.14 (0.23)	15.5%	2.3%	0.15 (0.23)	16.0%	1.2%
2-5yr Δ on lagged 5yr Δ	-0.38 (0.38)	3.2%	51.3%	-0.38 (0.37)	2.7%	49.9%

[a] at $p < 0.05$; Newey-West standard errors.

Extrapolation in Baseline Beliefs

	E(HPA), 1 year		E(HPA), 2-5 yrs (ann.)	
Past 1-yr Perceived HPA	0.226***		0.047***	
	(0.028)		(0.012)	
Past 5-yr Perceived HPA (annualized)		0.202***		0.080**
		(0.047)		(0.023)
E(fundamentals)	Y	Y	Y	Y
N	1020	1020	1020	1020
R-Squared	0.236	0.166	0.226	0.224

E(fundamentals) includes expectations of inflation, Δ(mortgage rate), Δ(rents), change in local economic conditions, and credit access.

- Significant positive correlations at individual level between perceptions and expectations, even after controlling for "fundamentals"
  $\Rightarrow$ suggestive of extrapolation
- But significantly weaker in L-frame than C-frame.

more

Results

	Δ E(HPA) at horizon: 1 year	2-5 year
T1	0.02	-0.12
	(0.29)	(0.11)
T5	0.10	0.10
	(0.29)	(0.11)
1yr Perception Gap	0.00	0.00
	(0.03)	(0.01)
5yr Perception Gap	0.05	0.00
	(0.05)	(0.02)
T1 × 1yr Perception Gap	0.20***	0.04**
	(0.04)	(0.02)
T5 × 5yr Perception Gap	0.07	0.05*
	(0.08)	(0.03)
Observations	1015	1013
R-Squared	0.056	0.024
Mean of dep. variable	0.29	-0.10
(St. dev.)	(3.04)	(1.47)

Heterogeneity in Updating Behavior

- Regression captures average behavior; may mask interesting heterogeneity
  - Some theories rely on heterogeneity, e.g. Malmendier-Nagel's learning-from-experience theory features age-dependent updating
- Study relationship with respondent characteristics in two stages (where $\Delta y =$ expectation revision):
  1. Share of non-updaters (relative to Control group respondents):

$$\frac{\text{\% of respondents in T1\&T5 with } \Delta y{=}0}{\text{\% of respondents in C with } \Delta y{=}0}$$

  2. Odds of an updating respondent being an Extrapolator (move in direction of gap) vs. Mean-reverter (move in opposite direction)

$$\frac{\text{\% of respondents in T1\&T5 with } (\text{Gap}\cdot\Delta y){>}0}{\text{\% of respondents in T1\&T5 with } (\text{Gap}\cdot\Delta y){<}0}$$

- Compare these quantities across subsamples, for both 1-year expectations and 5-year expectations
  - Multivariate analysis (appendix) shows similar patterns

Heterogeneity in Updating of 1-Year Expectations

	Sample Size	Non-Updater	p-value	Extrapolator/ Mean Reverter	p-value
Full Sample (T1+T5)	684	0.83		1.68	
T1	345	0.82	0.847	2.03	0.062
T5	339	0.83		1.40	
Age $<$ 40	202	0.90	0.483	2.56	0.045
Age 40-59	272	0.72		1.50	
Age $\geq$ 60	210	0.90		1.38	
Has Lived In Locality $<$ 15 years	332	0.87	0.563	1.90	0.247
Has Lived In Locality 15+ years	352	0.78		1.51	
No Negative Housing Market Experience	621	0.84	0.551	1.73	0.360
Foreclosure, Short Sale, or Underwater Mortgage	63	0.66		1.30	
Below-Median Supply Elasticity (Saiz 2010)	218	0.75	0.308	2.23	0.012
Above-Median Supply Elasticity	222	0.92		1.19	
Above-Median One-year Momentum	315	0.93	0.341	1.84	0.464
Below-Median One-year Momentum	304	0.79		1.58	

Heterogeneity in Updating of 5-Year Expectations

	Sample Size	Non-Updater	p-value	Extrapolator/ Mean Reverter	p-value
Full Sample (T1+T5)	684	0.77		1.72	
T1	345	0.90	0.004	1.86	0.446
T5	339	0.64		1.60	
Age $<$ 40	202	0.88	0.178	2.45	0.070
Age 40-59	272	0.86		1.43	
Age $\geq$ 60	210	0.63		1.58	
Has Lived In Locality $<$ 15 years	332	0.87	0.563	1.90	0.247
Has Lived In Locality 15+ years	352	0.78		1.51	
No Negative Housing Market Experience	621	0.77	0.653	1.86	0.014
Foreclosure, Short Sale, or Underwater Mortgage	63	0.86		0.88	
Below-Median Supply Elasticity	218	0.75	0.295	2.06	0.057
Above-Median Supply Elasticity	222	0.94		1.31	
Above-Median Long-term Mean Reversion	310	0.78	0.913	2.15	0.007
Below-Median Long-term Mean Reversion	309	0.80		1.25	

Heterogeneity – Takeaways

- Mixed evidence for age-dependent updating a la Malmendier-Nagel:
  - Young respondents not more likely to update
  - But they do extrapolate more (conditional on updating)
  - No relationship with length of tenure in town/city
- Respondents with negative experiences (e.g. past foreclosure) are the only sub-group who update in a way consistent with mean reversion for medium-term expectations
- Most intriguing: most extrapolation in places with lowest supply elasticity / strongest 5-year mean reversion
  - May be "rational" at short horizon (1yr)...
  - But not longer horizon!

Using behavior to forecast

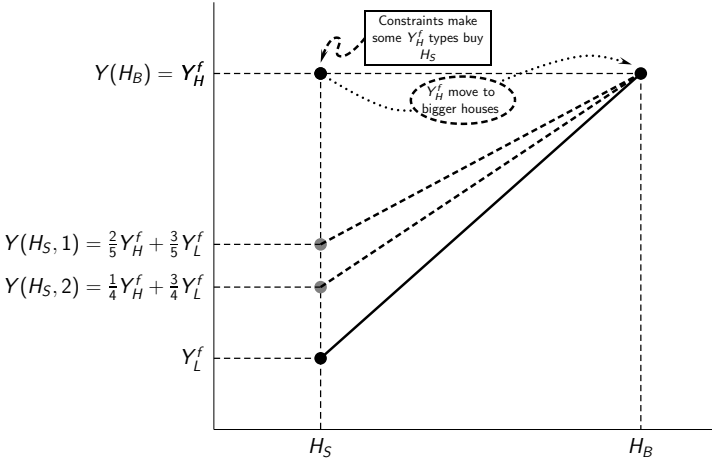
- Campbell (1987)
  - Permanent income says savings should forecast future income growth
  - High savings = low income growth
- Lettau and Ludvigson (2001)
  - High consumption to wealth ratio implies high future excess returns
  - Can use c/W to forecast excess returns on stocks.
- Gerardi, Rosen, and Willen (2010)
  - High spending on housing relative to income implies high expected income growth
  - Can use house spending to forecast income

Gerardi, Rosen, and Willen (2010)

- Two households

Household	Income $t=0$	Income $t=f$
High growth	y_0	Y_H^f
Low growth	y_0	Y_L^f

- Two possible houses
  - Big house costs H_B
  - Small house costs H_S.
- Assume households have different levels of wealth and if they could borrow unlimited amounts
  - $Y_H^f \Leftrightarrow H_B$
  - $Y_L^f \Leftrightarrow H_S$
- If we introduce constraints
  - $w > w^*$ Y_H^f buy H_B
  - $w \leq w^*$ Y_H^f buy H_S
  - All Y_L^f buy H_S



Basic Specification

- Regression Specification

$$\log y_{i,t+f} = \alpha_0 + \alpha_2 \log y_{it} + \alpha_3 \log H_{it} + \beta X_{it} + D_t + \alpha_4 * t \log H_{it} + \varepsilon_{it}$$

- Where
  - y_{it} is income at time of purchase
  - $y_{i,t+f}$ is income f years after purchase
  - H_{it} is purchase price of the house
  - X_{it} is demographic contols
  - D_t is a full set of time dummies
- Interact time and H_{it} to capture changes over time.

Dataset

- PSID: All households that purchased a home in the period spanning 1969-1999.
  - First-time homeowners who are transitioning from renting.
  - Existing homeowners who are moving into a new house.
- 2 variables that allow us to identify house purchases.
  - Tenure choice - renting or owning.
  - Moving variable - if family has moved in the past year.
- House variable - Self-reported value of principal residence (in real terms).
- Income variable - encompasses all real taxable labor and capital income for head of household and spouse (in real terms).
- Demographic variables - cubic in age, education (a series of dichotomous variables), race, sex of head, family size.

Main Results

- Future Income Definition - average of 2 and 4 years ahead.

(1) Left-hand side variable = avg. of 2 and 4 years ahead

		(1) Baseline	(2) Poverty	(3) First-Time Buyers	(4) Age < 40	(5) Age ≥ 40
(i)	log(House Value)	0.131***	0.155***	0.036	0.098***	0.186***
(ii)	Time$\times$ log(House Value)	0.004***	0.003*	0.006***	0.005***	0.002
(iii)	Breakpoint	1985***	1978*	1985	1981**	none
(iv)	log(House Value)	0.161***	0.141***	0.078***	0.126***	-
(v)	Dummy(Year$\geq$ Breakpoint)$\times$ log(House Value)	0.064***	0.089***	0.124***	0.076***	-
Observations		5,277	2,574	1,112	3,453	1,824

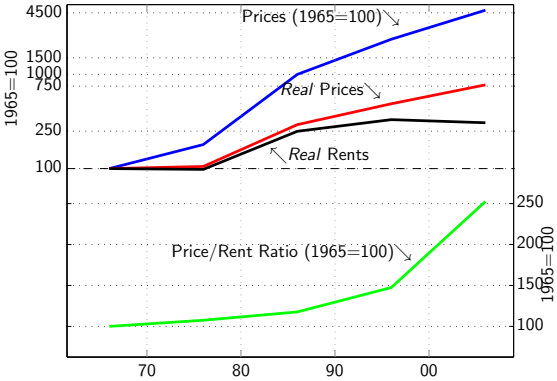
"Robustness"

- Results appear robust
  - House spending does not forecast future hours
  - House spending does not forecast occupational change
  - Works with both narrow earnings and broad household income
- Geographic controls (Not in current draft of the paper):
  - Full set of state dummies.
  - Full set of state-time interactions.
  - Within state regressions – small sample.

For changes

- What happened?
  - Creation of Secondary Market $\Leftarrow$ Candidate – timing is right
  - Expansion of mortgage choice set $\Leftarrow$ Candidate – timing is right
  - Credit Scoring $\Leftarrow$ Too late, early 1990s
  - Subprime Market $\Leftarrow$ Too late, mid-late 1990s
  - Increased individual level income volatility $\Leftarrow$ Timing is right, but people would buy smaller houses
  - Reduced economic volatility $\Leftarrow$ Timing is right, but effect goes the other way
  - End of Regulation Q $\Leftarrow$ Timing is right, but not clear if it mattered
  - Anti-discrimination laws $\Leftarrow$ Too early, but could explain poverty sample

Manhattan Real Estate



- What about rents?
- Real rents have gone up by about 3X.
- So Price-rent ratio has gone risen by $\sim 2.5X$
- All told, when we correct for inflation and rents.
- Increase goes from 50X to 2.5X.

Data Source: Miller Samuel

Is Manhattan Real Estate a Bubble?

- Gordon growth model

$$\frac{Price}{Rent} = \frac{1}{r - g}$$

- Up to now r and g, have been nominal variables

$$\frac{Price}{Rent} = \frac{1}{r_{\text{nom}} - g_{\text{nom}}}$$

- But, roughly speaking, since $r_{\text{nom}} = r_{\text{real}} + \pi$ and $g_{\text{nom}} = g_{\text{real}} + \pi$

$$\frac{Price}{Rent} = \frac{1}{r_{\text{nom}} + \pi - (g_{\text{nom}} + \pi)} = \frac{1}{r_{\text{real}} - g_{\text{real}}}$$

Is Manhattan Real Estate a Bubble?

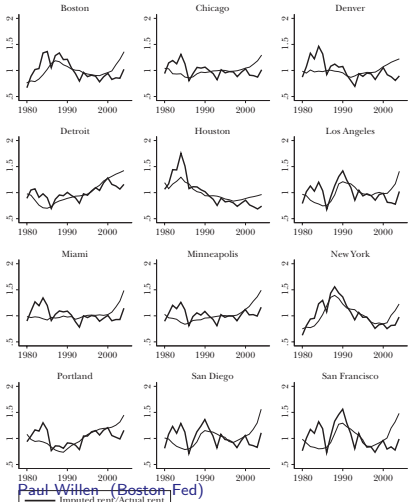
- Gordon Growth Formula

$$\frac{Price}{Rent} = \frac{1}{r_{\text{real}} - g_{\text{real}}}$$

- Why would P/R ratio go up by 2.5X?
- Real interest rates in 2005 were *higher* in 2005 than in 1965.
  - implying a lower P/R ratio...
- Maybe g went up What is g?
- Growth rate of rents
  - Expected – where do we find these expectations?
- To explain data,
  - Rent growth expectations must have gone way up...

Himmelberg, Mayer, and Sinai (2005)

Figure 2
Imputed-to-Actual-Rent Ratio versus Price-to-Rent Ratio
(ratios normalized to their 25-year average)



- Gordon Growth Formula:

$$\frac{\text{Price}}{\text{Rent}} = \frac{1}{r-g}$$

- Figure 2:

$$\underbrace{\text{Price} \cdot (r-g)}_{\text{Imputed Rent}} = \underbrace{\text{Rent}}_{\text{Actual Rent}}$$

  - Use historic rent growth for expected rent growth.

- Alternative formulation

$$\underbrace{g}_{\text{Historic Growth}} = \underbrace{r - \frac{\text{Price}}{\text{Rent}}}_{\text{Forecast Growth}}$$

Adaptive expectations

- JPMorgan "HPA Update"

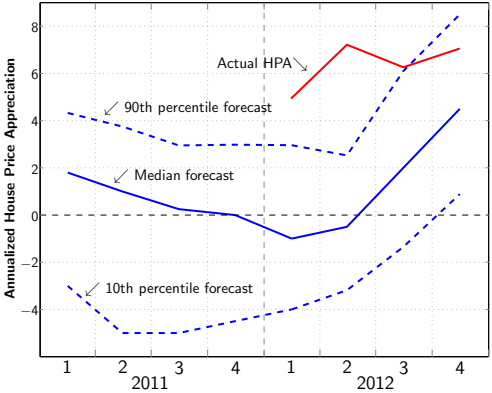
Date of	Data from	Title
12/8/06	10/06	"More widespread declines with early stabilization signs"
1/10/07	11/06	"Continuing declines with stronger stabilization signs"
2/6/07	12/06	"Tentative stabilization in HPA"
3/12/07	1/07	"Continued stabilization in HPA"
9/20/07	7/07	"Near bottom on HPA"
11/2/07	9/07	"UGLY! Double digit declines in August and September"

- JP Morgan, May 16, 2008.: "We expect another 15% drop in home prices over the next 12 months."

House Prices

- Since 2010, Zillow.com has conducted a survey of economist about their expectations for house prices.
- A Who's Who of housing economics
  - Academics
  - Bankers
  - Builders
  - Policy Makers
- How have they done?

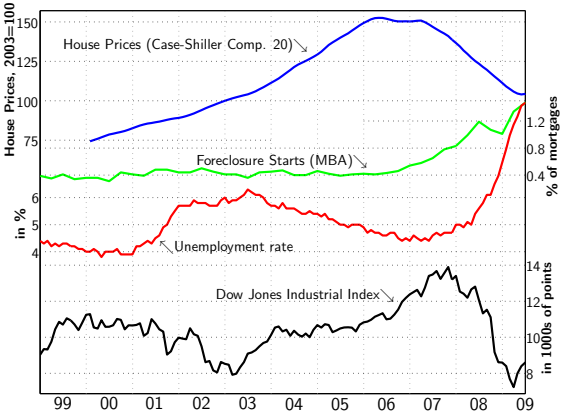
Forecasts of 2012 House Prices



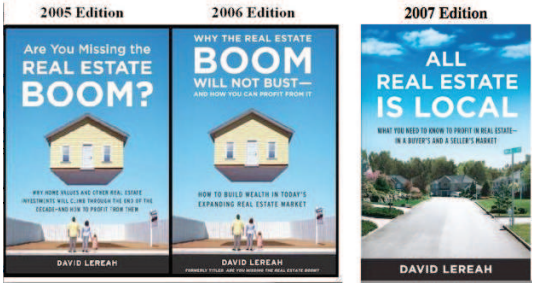
- More "Adaptive Expectations"

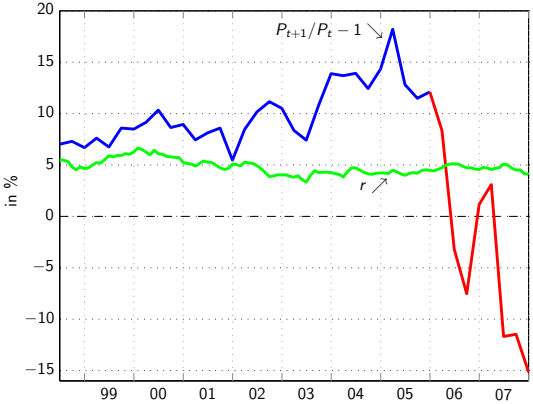
Gerardi, Foote, and Willen (2011)

- Second most destructive bubble in U.S. history
- Burst sometime in 2006
- No Black Tuesday, but by 2007, it was clearly a disaster.



Housing Optimists: David Lereah





- Made all kinds of arguments.
  - Finite amount of land
  - Population growth
  - "Everyone always needs a place to live.
- But the real argument was...
  - EMH is wrong, look at the data.

Investment banks

- Consensus was that decline on the order we have seen was highly unlikely.
- No national decline in history...

  ...the risk of national decline in home prices appears remote. The annual HPA has never been negative in the United States going back at least to 1972.

- Regional falls:

  In each one of these regional corrections, the decline of home prices coincided with a deep regional recession.

- "few economists predict a near-term recession in the U.S." (Citi, December 2, 2005.)

Housing Pessimists: No explanation for the surge

In the absence of any other credible theory, the only plausible explanation for the sudden surge in home prices is the existence of a housing bubble. This means that a major factor driving housing sales is the expectation that housing prices will be higher in the future. While this process can sustain rising prices for a period of time, it must eventually come to an end.

Dean Baker (2002)

- Dean Baker dismissed all the possible covariates as unconvincing.
- In 2002.
- House prices are higher today than in 2002...

Pessimists: John Paulson

- Made billions betting against subprime loans.
- Did not focus on the loans – everyone knew they were bad.
- Believed that the key to mortgage defaults was house prices:

> *Paulson and Pellegrini concluded that the only way their trades would work was if the U.S. real estate market had reached unsustainable levels and began to fall..."*

Housing Pessimists: Paulson

- Big insight – "Eureka" moment.

  > *Housing prices had climbed a puny 1.4 percent annually between 1975 and 2000, after inflation was taken into consideration. But they had soared over 7 percent in the following five years, until 2005. The upshot: U.S. home prices would have to drop by almost 40 percent to return to their historic trend line. (Zuckerman, loc 1936).*
  >
  > *The chart [showing that house prices would fall 40 percent] was Paulson's Rosetta stone, the key to making sense of the entire housing market.*

- Does this make sense?

Housing Agnostics: Mean reversion

- Researchers (mostly in the Fed system...) essentially did the scientific equivalent of looking at the picture: estimated time series models.
- Davis, Lehnert and Martin (2008 but written in 2006)
  - From 1960-1995, the rent-price ratio is fairly constant
  - This ratio declines dramatically from 1995-2006
- The bottom line?

  If the rent-price ratio were to rise from its level at the end of 2006 up to about its historical average value of 5 percent by mid-2012, house prices might fall by 3 percent per year, depending on rent growth over the period. That said, this is more of a back-of-the-envelope calculation than an actual forecast for house prices because we do not have a fully satisfactory model of the rent-price ratio.

Agnostics: Shiller in 1996...

- Shiller predicted low returns on stocks after Greenspan's "irrational exuberance" comment in 1996.

> *The conclusion of this paper that the stock market is expected to decline over the next ten ears and to earn a total return of just about nothing has to be interpreted with great caution.*
>
> *Our search over economic relations that led us to study the price divided by 30-year moving average of earnings may have stumbled upon a chance relation with no significance.*
>
> *It is also dangerous to assume that historical relations are necessarily applicable to the future. There could be fundamental structural changes occurring now that mean that the past of the stock market is no longer a guide to the future.*

Models of Expectations

1. Rational Expectations (Muth, 1961)
   - "Nobody goes there any more. It's too crowded."
   - "The reason that the oil price is going to stay high is that no one is investing in new capacity because they think the price is going to fall."
2. Adaptive Expectations (Nerlove, 1958)
3. Natural Expectations (Fuster, Laibson, and Mendel, 2010; Fuster, Hebert, and Laibson, 2012)
4. Extrapolative Expectations (Negishi, 1964; Glaeser and Nathanson, 2015)
5. Epidemiological models (Burnside, Eichenbaum, and Rebelo, 2016)

Burnside, Eichenbaum, and Rebelo (2016)

- Model:
  - linear utility discounted at β.
  - continuum of households [01]
  - continuum of owner occ. houses $[0k]$ $k < 1$ selling at price P_t generating utility ε^h
  - continuum of renter houses perfectly elastically supplied at rent w generating utility ε^r
  - no short sales *and* only one house per person (no long sales).
- Pricing condition:

$$-P_t + \beta(P_{t+1} + \varepsilon^h) = \beta(\varepsilon^r - w)$$

- Steady state

$$P_{ss} = \frac{\beta}{1-\beta}(\varepsilon^h - (\varepsilon^r - w)) = \frac{\beta}{1-\beta}\varepsilon$$

where $\varepsilon \equiv \varepsilon^h - (\varepsilon^r - w)$

Burnside, Eichenbaum, and Rebelo (2016) II

- At time 0, all households learn that with probability ϕ, ε will change to ε^*
- Three types of agents
  1. Optimists (o): $\mathrm{E}^o(\varepsilon^*) > \varepsilon$
  2. Skeptics (s): $\mathrm{E}^s(\varepsilon^*) = \varepsilon$
  3. Vulnerable (v): $\mathrm{E}^v(\varepsilon^*) = \varepsilon$
- Suppose there are only o and s and $o > k$ so o is the marginal buyer:

$$P_t^o = \beta\{\phi[\mathrm{E}^o(\varepsilon^*) + (1-\phi)(\varepsilon + P_{t+1}^*)]\}$$

- Steady state

$$P^o = \beta \frac{\phi \mathrm{E}^o(\varepsilon^*)/(1-\beta) + (1-\phi)\varepsilon}{1-\beta(1-\phi)}$$

- Pricing condition:

$$-P_t + \beta(P_{t+1} + \varepsilon^h) = \beta(\varepsilon^r - w)$$

- Steady state

$$P_{ss} = \frac{\beta}{1-\beta}(\varepsilon^h - (\varepsilon^r - w)) = \frac{\beta}{1-\beta}\varepsilon^*$$

where $\varepsilon^* \equiv \varepsilon^h - (\varepsilon^r - w)$

The slide you've all been waiting for...

- The end.

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